

Financial Machine Learning: Data Structures, Advanced Labeling & Statistical Foundations

AI Finance Seminar

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Chapter 2 Overview

The Data Pipeline Problem

Raw financial data (tick-by-tick transactions) is highly irregular, noisy, and non-IID. We must transform this unstructured flow into structured data bars optimal for machine learning algorithms.

Structure of this Section:

- **Textbook Background:** Volatility Clustering, Microstructure & Corporate Actions
- **Standard Bars:** Time, Tick, Volume, and Dollar Bars
- **Information-Driven Bars:** Imbalance and Runs Bars
- **Multi-Product Series:** The ETF Trick
- **Event-Based Sampling:** The CUSUM Filter
- **Simulation Demo:** Quantifying Normality Recovery

Ch.2 Background 1: Volatility Clustering & Time Bars

Academic vs. Empirical Time

Standard econometrics samples data at equal intervals of chronological time (e.g., daily, hourly). However, markets do not process information at a constant chronological pace.

- **Volatility Clustering (Heteroscedasticity):** Financial markets exhibit structural variations in activity. High-volatility periods cluster together, followed by quiet periods.
- **Intraday Seasonality (The U-Shape):** Volume and volatility spike near market Open and Close, while drying up during mid-day.
- **The Consequence:** Sampling by constant chronological intervals forces us to over-sample quiet periods (adding noise) and under-sample volatile periods (missing key information). This violates the IID normal distribution assumption required by ML.

Ch.2 Background 2: Market Microstructure & Order Book

- **The Limit Order Book (LOB):**

- **Bids** represent buyers' resting limit orders; **Asks** represent sellers' resting limit orders.
- **Bid-Ask Spread:** The friction cost of immediate execution. Crossing the spread (buying at Ask, selling at Bid) represents an immediate transaction cost.

- **Order Fragmentation:**

- Institutional algorithms split massive trades into thousands of micro-orders (ticks) to hide their market impact.
- This means **Tick counts** are heavily manipulated by trading systems and do not accurately represent the true economic volume exchanged.

Ch.2 Background 3: Corporate Actions & Price Adjustments

- **Corporate Actions:**

- **Stock Splits / Reverse Splits:** Artificially change nominal stock prices and outstanding shares without altering company equity.
- **Dividends:** Cash distributions that instantly drop the stock price on the ex-dividend date.

- **Data Adjustment and Its Pitfall:**

- Back-adjusting historical prices is necessary but can introduce **Look-ahead Bias** if future adjustment factors are accidentally leaked into historical trading simulations.
- Standard Volume and Tick bars are vulnerable to sudden jumps during splits, whereas **Dollar Bars** (tracking total currency exchanged) adjust dynamically and remain stable.

Topic 2.1: The Four Essential Data Types

To feed our ML models, we must understand the taxonomy of financial data:

Data Type	Description	ML Challenges
Fundamental	Balance sheets, earnings, macro indicators	Low frequency, revisions, lag
Market	L3/L2 Order book feed, executed ticks	High frequency, noise, size
Analytics	Sentiment scores, news signals, analyst ratings	Highly subjective, sparse
Alternative	Satellite imagery, credit cards, shipping data	High cost, unstructured, short histo

Topic 2.2: Standard Data Bars (Why Use Dollar Bars?)

Standard bars aggregate tick-by-tick transactions based on a fixed threshold:

- **Time Bars:** Sampled every t minutes. Suffer from heavy non-normality.
- **Tick Bars:** Sampled every N transactions. Captures transaction counts, but distorted by order splitting.
- **Volume Bars:** Sampled every V shares/contracts traded. Captures size, but ignores nominal value.
- **Dollar Bars (The de Prado Recommendation):**
 - Sampled every D currency units (e.g., \$10M) exchanged.
 - **Self-Adjusting:** As price inflation or splits occur, the share count automatically adjusts, maintaining a stable sampling rate.
 - **Statistical Superiority:** Dollar-bar returns exhibit distributions much closer to a Gaussian normal distribution with stable variance (homoscedasticity).

Topic 2.3: Information-Driven Bars (Imbalance Concept)

Informed Traders and Order Flow

Informed traders possess non-public or superior analysis, causing them to trade aggressively in one direction, creating order flow imbalance.

Let $b_t \in \{-1, +1\}$ be the trade sign via the tick rule:

$$b_t = \begin{cases} b_{t-1} & \text{if } \Delta p_t = 0 \\ \frac{|\Delta p_t|}{\Delta p_t} & \text{if } \Delta p_t \neq 0 \end{cases}$$

The cumulative imbalance over a bar of length T is $H_T = \sum_{t=1}^T b_t$ (or weighted by volume/dollars).

Imbalance Bars Condition

Sample a bar when the cumulative imbalance exceeds its dynamically updated expectation:

$$|H_T| \geq E_0[T \cdot |2P[b_t = 1] - 1|]$$

Topic 2.3: Information-Driven Bars (Runs Concept)

While imbalance bars look at the net signed flow (which can cancel out), **Runs Bars** look at the absolute length of consecutive buys or sells. They are designed to capture institutional algorithms that split large orders into consecutive runs. Let

$$\theta_T = \max \left(\sum_{t|b_t=1} b_t, - \sum_{t|b_t=-1} b_t \right).$$

Runs Bars Condition

We sample a bar when the run sequence of a single direction exceeds expected parameters:

$$\tau \geq E_0[T]$$

Topic 2.4: Multi-Product Series (The ETF Trick)

The Practical Challenge

ML models cannot easily predict multiple raw price series that have differing contract sizes, transaction costs, rollovers, and weightings (e.g., spread trading or physical portfolios).

- **The ETF Trick:**

- A rigorous mathematical framework that models a complex, dynamic basket of assets (with rebalancing, dividend reinvestment, and funding costs) as a single, continuous, investable 1D price series.
- Incorporates transaction costs (crossing the bid-ask spread) at each rebalancing.
- This creates a clean, tradeable representation of a trading strategy's equity curve, which can then be predicted by ML models without look-ahead bias.

Topic 2.5: Event-Based Sampling (The CUSUM Filter)

To avoid over-fitting and reduce computational cost, we should only feed the ML model data points that represent a **genuine structural event** or price breakthrough.

Symmetric Cumulative Sum (CUSUM) Filter

An event-detection scheme that identifies local regime shifts in a log-price series y_t :

$$S_t^+ = \max(0, S_{t-1}^+ + y_t - y_{t-1} - e)$$

$$S_t^- = \min(0, S_{t-1}^- + y_t - y_{t-1} + e)$$

An event is sampled when $\max(S_t^+, -S_t^-) \geq h$, where h is a threshold (often set as a function of rolling volatility) and e is a drift tolerance. After sampling, the filters are reset.

Empirical Demonstration: Time Bars vs. Dollar Bars

Statistical Summary of Normality Recovery

By aligning the sampling clock with transaction value flow rather than physical time, Dollar Bars suppress intraday heteroscedasticity, producing near-Gaussian returns.

Sampling Method	Skewness	Excess Kurtosis	Jarque-Bera p-val	Normality Assum
Time Bars (10-Min)	-0.12	1.85	< 0.0001	Strongly Rejected
Dollar Bars (Matched N)	+0.02	0.20	0.8473	✓ Accepted (Near

Note: Realized excess kurtosis is cut by approximately 89% using Dollar Bars, resolving tail risks for ML models.

Python Code: Simulating Normality Recovery

Extract from normality-simulation-demo.py

```
# Calculate Returns
time_returns = np.diff(np.log(time_bars['price']))
dollar_returns = np.diff(np.log(dollar_bars['price']))

# Test Normality
time_jb = stats.jarque_bera(time_returns)
dollar_jb = stats.jarque_bera(dollar_returns)

print(f"Time Bars -> Kurtosis: {stats.kurtosis(time_returns):.2f}, JB p-val: {time_jb.pvalue}")
print(f"Dollar Bars -> Kurtosis: {stats.kurtosis(dollar_returns):.2f}, JB p-val: {dollar_jb.pvalue}")
```

Chapter 3 Overview

The Target Problem

A machine learning model cannot predict "prices." It must predict structured labels (y) representing investment outcomes. How we define these labels determines whether our model is tradeable or merely a mathematical illusion.

Structure of this Section:

- **Textbook Background:** Stop-Loss/Profit-Taking, Path-Dependency & Bet Sizing
- **Traditional Labeling:** The Fixed-Time Horizon Method and its Flaws
- **Dynamic Thresholds:** Volatility-adjusted Barriers
- **The Triple-Barrier Method:** Modeling Real Trading Rules
- **Meta-Labeling:** Decoupling Trade Direction (Side) from Bet Size
- **Class Imbalance Mitigation:** Dropping Unnecessary Labels

Ch.3 Background 1: Trading Rules & Path-Dependency

European vs. Barrier Options Analogy

Traditional finance research treats investment labels like European options (evaluating outcomes only at a fixed end-point). Real-world trading, however, behaves like Barrier options.

- **Profit-Taking (Limit) & Stop-Loss Rules:** Every institutional trade is bound by strict risk management boundaries. If a position drops too far, it is immediately closed (Stop-loss). If it hits the target, profits are realized (Profit-taking).
- **Path-Dependency:** The path of the price matters. Even if a stock ends up 20% higher in 30 days, if it dipped 50% on day 3, a real-world trader would have been liquidated. We must use **path-dependent labeling** to account for this.

Ch.3 Background 2: Bet Sizing & The Kelly Criterion

Decoupling Direction from Bet Sizing

Finding a profitable opportunity (Is the market going up or down?) is only half the battle. Surviving the market requires deciding **how much to bet** (Bet Sizing).

- **The Danger of Uniform Betting:** Betting the same amount on every signal regardless of model confidence leads to rapid bankruptcy during a string of bad luck.
- **The Kelly Criterion:** A classical formula that dictates the optimal bet size based on the probability of winning (p) and the win/loss ratio (b):
$$f^* = \frac{p \cdot b - (1-p)}{b}$$
- **The Role of ML:** We need our labeling system to feed the ML model not just the direction, but the **probability of success** to feed into our bet sizing algorithm.

Ch.3 Background 3: ML Metrics vs. Investment Metrics

The Misalignment

A machine learning engineer optimizes for **Accuracy**. A portfolio manager optimizes for the **Sharpe Ratio** and minimum drawdown. These are not always aligned.

- **The Fallacy of High Accuracy:** A model with 90% accuracy can still bankrupt a fund if its 10% failures are catastrophic drawdowns (large losses, small gains).
- **Precision vs. Recall in Finance:**
 - **Recall (Opportunity capture):** "Out of all profitable trades, how many did we take?"
 - **Precision (Quality of bets):** "Out of all trades we took, how many were actually profitable?"
- **F1-Score to Sharpe Ratio:** Maximizing the F1-Score (balance of Precision & Recall) directly translates to a higher Sharpe Ratio by reducing transaction costs and eliminating false positives.

Topic 3.1: The Fixed-Time Horizon Method & Its Flaws

The traditional academic approach: $y_i = 1$ if $P_{i+h} > P_i + \tau$, else -1 or 0 . **Why this fails in practice:**

- **Constant Threshold Illusion:** A fixed threshold (e.g., $\tau = 1\%$) in a low-volatility regime will produce only 0 labels. In a high-volatility regime, it will produce random ± 1 labels. This breaks stationary distribution properties.
- **No Intra-horizon Stops:** It assumes the portfolio manager will hold the asset blindly until time h , completely ignoring stop-loss or profit-taking events that would have occurred within the horizon.

The Solution

We must make the profit/loss thresholds **dynamic** (based on rolling volatility) and the labeling **path-dependent** (tracking intra-horizon price action).

Topic 3.2 & 3.3: The Triple-Barrier Method

To model realistic trading, de Prado introduces the **Triple-Barrier Method**. We define three barriers:

- **Barrier 1 (Upper Horizontal):** Profit-taking limit ($\theta_1 \cdot \sigma_t$)
- **Barrier 2 (Lower Horizontal):** Stop-loss limit ($-\theta_2 \cdot \sigma_t$)
- **Barrier 3 (Vertical):** Maximum holding time (expiration) ($t + H$)

The label y_i is determined by which barrier is touched first:

$$y_i = \begin{cases} 1 & \text{if Upper Barrier is touched first} \\ -1 & \text{if Lower Barrier is touched first} \\ 0 \text{ (or sign of return)} & \text{if Vertical Barrier is touched first} \end{cases}$$

Dynamic Volatility Adjustment

σ_t is the rolling exponentially-weighted volatility of returns. This scales the barriers dynamically, ensuring the model's target labels adapt to changing market regimes.

Topic 3.4: Meta-Labeling (Decoupling Side & Size)

Two-Stage Learning Architecture

Instead of asking one ML model to determine both **direction** and **bet size**, we split the problem into a primary and secondary model.

① Primary Model (Primary Signal):

- Can be a technical rule (e.g., SMA crossover) or a fundamental signal.
- Determines the **Side** ($s_i \in \{-1, +1\}$: Buy or Sell).

② Secondary Model (Meta-Labeling):

- We apply the Triple-Barrier Method **only in the direction** of the primary signal.
- If the trade hits profit-taking, the label is $y_i^{meta} = 1$ (True Positive).
- If the trade hits stop-loss, the label is $y_i^{meta} = 0$ (False Positive).
- The Secondary ML model learns to predict the **probability of success** ($P[y^{meta} = 1]$).

The Superpowers of Meta-Labeling

Why is Meta-Labeling the preferred architecture of quantitative hedge funds?

- **Mitigates Class Imbalance:** The secondary model only evaluates trades when a primary signal occurs. It focuses purely on filtering bad trades (optimizing precision).
- **Enables "Quantamental" Trading:**
 - Keeps the primary model as a clear, intuitive, economically-justified "white box" (e.g., MACD or fundamental value).
 - Overlays a sophisticated ML "black-box" as a risk filter (secondary model) to size the bet.
- **Prevents Overfitting:** Since the ML model does not decide the direction (which is highly noisy), but only whether to execute and how much to bet, the model's variance and overfitting risks are minimized.

Graduate Advanced Overview

The Deep Statistical Reality

To achieve mathematical rigor at the PhD/Graduate level, we must resolve two fundamental violations of classical probability theory in financial machine learning.

- 1 **The Stationarity vs. Memory Dilemma:** Standard price series are non-stationary, but standard return series (differencing) erase all long-term historical memory.
- 2 **Non-IID Sample Overlap:** Overlapping labels in the Triple-Barrier Method lead to highly correlated training observations, leading to Bagging (Random Forest) failure.
- 3 **Empirical Covariance Instability:** Raw empirical covariance matrices are dominated by noise, destroying Markowitz portfolio optimization.

Topic 4.1 & 4.2: Fractional Differentiation (FFD)

To make a series X_t stationary, standard practice is integer differencing $d = 1$ (calculating returns). However, this wipes out the long-term historical memory needed for predictive modeling.

Fractional Differentiation ($0 < d < 1$)

Using binomial expansion, we generalise the differencing operator to non-integers:

$$(1-B)^d X_t = \sum_{k=0}^{\infty} (-1)^k \binom{d}{k} B^k X_t = \sum_{k=0}^{\infty} \omega_k X_{t-k}$$

where $\omega_0 = 1$ and $\omega_k = -\omega_{k-1} \frac{d-k+1}{k}$.

- **FFD (Fixed-Width Window):** We drop terms where $|\omega_k| < \tau$ (a precision tolerance, e.g., 10^{-4}) to keep a constant lookback window.
- **The Benefit:** FFD finds the exact minimum d^* that achieves stationarity (ADF test passed) while preserving over 90% of the original price's correlation and long-term memory.

Topic 4.3: Sequential Bootstrap (Label Uniqueness)

- **The Concurrency Problem:** If a triple-barrier label spans from t_1 to t_2 , and another spans from $t_1 + 1$ to $t_2 + 1$, they share the same underlying market returns. They are not independent.
- **Failure of Standard Bagging:** Uniform bootstrap sampling with replacement assumes IID data. Under overlapping labels, it draws highly redundant samples, leading to massive out-of-sample over-fitting.

The Sequential Bootstrap Solution

Instead of uniform probability, we dynamically calculate the ****uniqueness**** of a sample i relative to the already drawn sample set:

- ① Compute the concurrency of active labels at each timestamp.
- ② Calculate the uniqueness $u_i^{(t)}$ of each observation.
- ③ Sample the next row with a probability proportional to its uniqueness.

This restores the independence assumption of ensemble models, unlocking the true power of Random Forests in finance.

Topic 4.4: Random Matrix Theory (Denoising Covariances)

Empirical covariance matrices Σ from asset returns are highly unstable and noisy.

The Marcenko-Pastur Theorem

For a random matrix with dimensions $T \times N$, if $T, N \rightarrow \infty$ with $1 < \frac{N}{T} = q < \infty$, the eigenvalues λ of its correlation matrix follow:

$$f(\lambda) = \frac{q}{2\pi\sigma^2\lambda} \sqrt{(\lambda_+ - \lambda)(\lambda - \lambda_-)} \quad \text{for } \lambda \in [\lambda_-, \lambda_+]$$

where $\lambda_{\pm} = \sigma^2(1 \pm \sqrt{1/q})^2$. Eigenvalues within $[\lambda_-, \lambda_+]$ represent pure noise.

• Denoising via Constant Residual Eigenvalues:

- Keep eigenvalues $\lambda > \lambda_+$ (Signal).
- Replace all eigenvalues $\lambda \leq \lambda_+$ (Noise) with their average value, keeping the trace (total variance) constant.

Topic 4.4: Random Matrix Theory (Detoning)

The "Market Component"

The largest eigenvalue λ_{max} of a financial correlation matrix is typically massive, representing the market-wide dominant factor (all assets moving together).

- **The Detoning Process:**

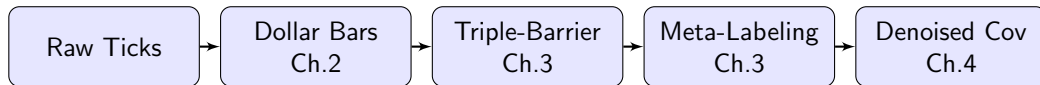
- Financial clustering (unsupervised classification) can be blinded by this dominant market-wide component.
- Detoning removes the first eigenvector (associated with λ_{max}) from the correlation matrix:

$$\mathbf{C}_{detoned} = \mathbf{C} - \lambda_1 \mathbf{W}_1 \mathbf{W}_1^T$$

where $\mathbf{W}_1$ is the first eigenvector.

- **Result:** This allows algorithms to focus on sector-specific, micro-alpha relationships and subtler corporate clusters without market-wide bias.

Summary: The Complete Financial ML Pipeline



Main Takeaway

Successful financial ML is not about finding a complex algorithm. It is about building a mathematically rigorous data and labeling pipeline that respects market microstructure, path-dependency, and statistical boundaries.

Q&A

Thank You! Questions & Discussions

AI Finance Seminar, Sogang University